Optimal Serve Placement in Professional Tennis Accounting for Execution Error

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Abstract

We study where tennis players should aim their serves conditional on their execution error. Aiming closer to the lateral court boundaries increases the difficulty of the return; however, it also increases the risk of faulting since no player can execute their serve perfectly. Optimal targets need to balance this tradeoff, accounting for both player execution error and the two-period nature of a point in tennis. Using spatial serve data from the 2019-2021 Australian Open professional tennis tournaments, we first fit a Generalized Additive Mixed Model (GAMM) to quantify the relationship between a serve's bounce location and the probability of winning the point. We then fit a 3-component bivariate Gaussian Mixture Model (GMM) to model each player's distribution of serves. Using the conditional point win probabilities provided by the GAMM and the serve execution densities provided by the GMM, we estimate the expected point value of aiming at each location over a fine grid of possible targets within the returner's half of the court. We then estimate optimal targets for each player, conditional on serve period and strategy (Wide or T). We compare player's estimated targets from the data with the optima we prescribe, finding that players tend to aim more conservatively than optimal, leaving room to target closer to the lateral boundaries of the service box. Lastly, we find a positive association between aiming closer to our prescribed optima and winning more points.

1 Introduction

The serve is the most consequential shot for winning or losing a point in tennis. A well-executed serve puts the returner in a difficult position, increasing the chances of an ace or a weak return. On the other hand, a poorly-executed serve can lose the server the point by faulting or by giving the returner an opportunity for a strong return. The strength of the serve and subsequent return has ripple effects throughout an entire rally, highlighting the importance of serving well. The serve is also the shot that a player has the most control over, since the ball starts in their hand rather than needing to be intercepted. Therefore, the most consequential shot in a rally is also the most practical to optimize.

When serving, a player must land their serve in the diagonally opposite service box, which is bounded by the net, service line, center service line, and the court-specific singles sideline, as shown in Figure 1. If the 1st serve attempt is valid, then play continues until someone wins the point. If not, then it is referred to as a "fault", and no point is awarded. The server then gets a 2nd attempt, and if they fault again, the returner wins the point.

There are three main serve strategies. The first is aiming "out wide" (Wide), which means serving towards the singles sideline. The second is aiming "up the T" (T), meaning along the center

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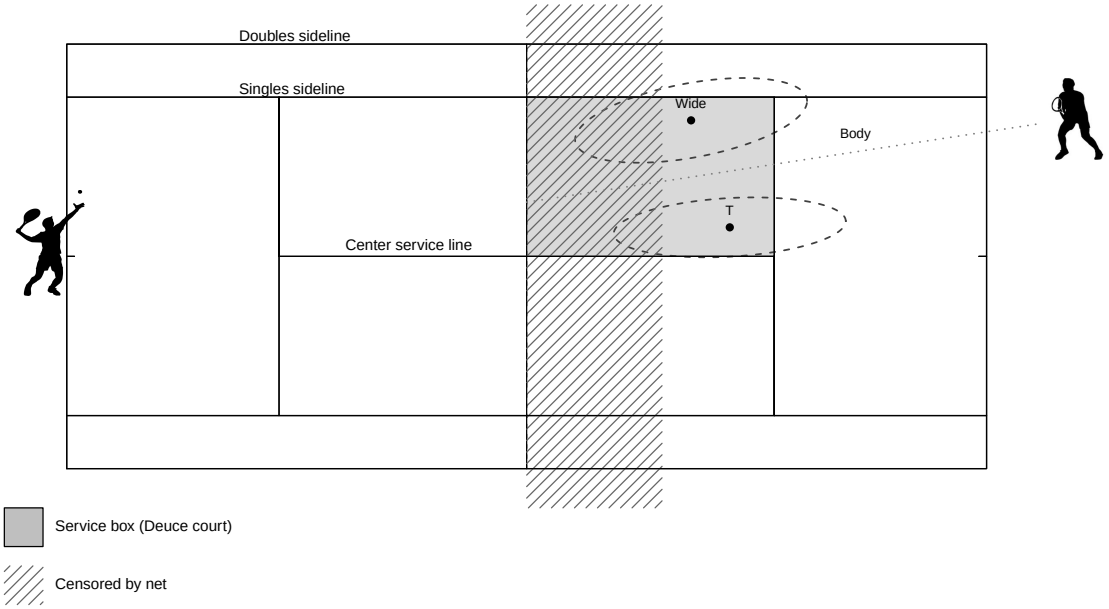


Figure 1: Illustration of the serving decision for a Deuce court serve, reproduced from Sandholtz et al. [2026]. Dashed ellipses represent the server's execution error around an intended target.

service line. The third strategy involves serving into the returner's body, jamming them between their forehand and backhand, possibly forcing an awkward return. A server must mix between these strategies in order for each of them to remain effective. The optimal mixture between these strategies has been studied previously (e.g., Walker and Wooders [2001]).

Sandholtz et al. [2026] studied the optimal aiming problem *conditional on a given strategy* (i.e., conditional on aiming "out wide" or "up-the-T"). In this context, the server faces a natural risk vs reward tradeoff when deciding how far to aim their serve from the returner's position. The closer the server aims towards the singles sideline (for wide serves) or the center service line (for T serves), the further the returner will have to run and the harder it will be for them to return. However, aiming closer to the boundaries increases the risk of faulting by accidentally landing a serve outside the service box. On the other hand, hedging in from the service court boundaries decreases the risk of faulting, but also gives the opponent an advantage on their return. Optimal aiming must balance this tradeoff.

The optimal serve target depends on a server's accuracy. A server with perfect accuracy can always aim as far away from the returner as possible, since it maximizes the likelihood of an ace with no risk of faulting. However, the greater the execution error of the player, the further they may have to hedge-in from service box boundaries. Optimal targets are therefore player specific. In addition, optimal targets depend on the serve attempt number, since faulting carries a significantly greater penalty on 2nd serve than on 1st.

One limitation of Sandholtz et al. [2026] is that the optimal targets they estimate cannot be empirically validated. This is because their analysis was based on serve data collected from collegiate players in an experimental setting where no points were played out. In this analysis, we implement the modeling framework of Sandholtz et al. [2026] on observed match-play data, and show that

aiming closer to the prescribed optima is associated with a greater point win probability.

The dataset we use comes from the 2019-2021 Australian Open tennis tournaments. It includes the spatial bounce location of each serve in these tournaments, along with the server name, returner name, serve number, court side, who won the point, and other contextual variables. An example plot of Novak Djokovic's serve data is shown in Figure 2.

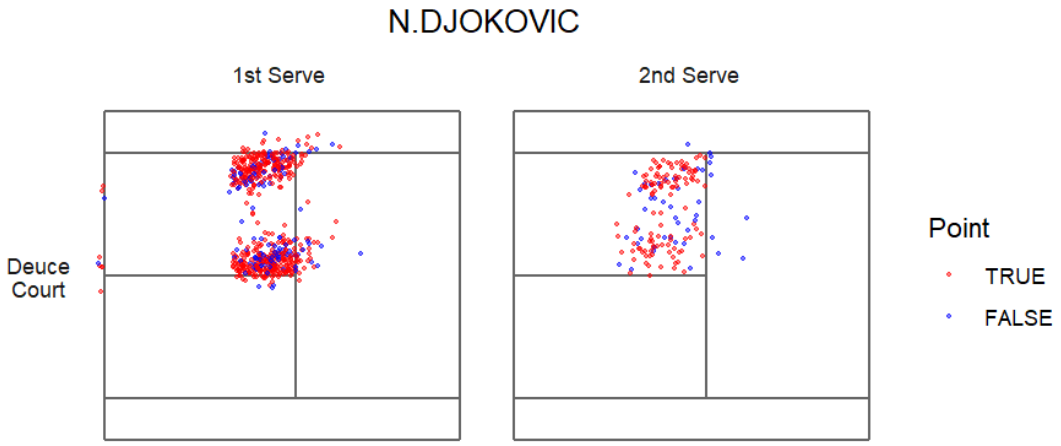


Figure 2: Bounce locations of Novak Djokovic's Deuce court serves from the 2019-2021 Australian Open tournaments. Points are colored according to whether he won the point. Note that servers can still win the point even if they fault on 1st serve.

The remainder of the paper proceeds as follows. We first model the relationship between serve landing location and point win probability (Section 2). We then estimate each player's execution error and inferred aiming locations (Section 3), and combine these estimates to identify optimal targets for first and second serves within each Wide and T region (Section 4). Finally, we compare players' inferred targets with the prescribed optima and examine whether aiming closer to these optima is associated with a higher probability of winning the point (Section 5).

2 Reward Surface

In order to solve the optimal aiming problem, we need a surface of the probability that the server wins a point conditional on a serve's bounce location. We model this surface using the Australian Open data via the bounce coordinates of each serve along with who won the point. The two-dimensional bounce coordinates represent the main explanatory variables of interest, while an indicator for whether the server won the point serves as a binary response. We do not model serves which land outside the service boxes (i.e., faults) since the win probability in these regions can be manually defined.

The covariates that will be included in our model are:

- x, y: Two-dimensional bounce location of a serve in meters.
- s: Serve speed (kph).
- Server/Returner Identifiers.

- Hand-Court: Combination of server handedness and court side. The possible values this can take are "Cross Body", which includes right-handed Deuce court serves and left-handed Ad court serves, and "Same Side", which includes left-handed Deuce court serves and right-handed Ad court serves.

We fit a Generalized Additive Mixed Model (GAMM) with a Bernoulli response akin to the point outcome model of Kovalchik et al. [2020]. Specifically,

$$Point_i \sim \text{Bernoulli}(p_i) \tag{1}$$

$$\begin{aligned}\text{logit}(p_i) = \alpha + \beta \cdot \mathbf{1}[Hand\text{-}Court_i = \text{"Cross Body"}] + f_{Hand\text{-}Court_i}(x_i, y_i, s_i) \\ + u_{\text{server}[i]} + u_{\text{returner}[i]}\end{aligned} \tag{2}$$

where

$$u_{\text{server}[i]} \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2_{\text{server}})$$
$$u_{\text{returner}[i]} \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2_{\text{returner}})$$

are random intercepts for the server and returner, respectively, and where $f_{\text{CrossBody}}$ and f_{SameSide} are separate thin plate regression splines in the joint space of bounce location and serve speed.

After fitting the model, we can generate a win probability surface on a fine grid over the Ad and Deuce service boxes for each serving player and serve period. In doing so, we use a player's average serve speed for each serve period, and we also assume a tournament average returner (i.e., we do not plug in a returner random effect). These surfaces are visualized for Djokovic in Figure 3.

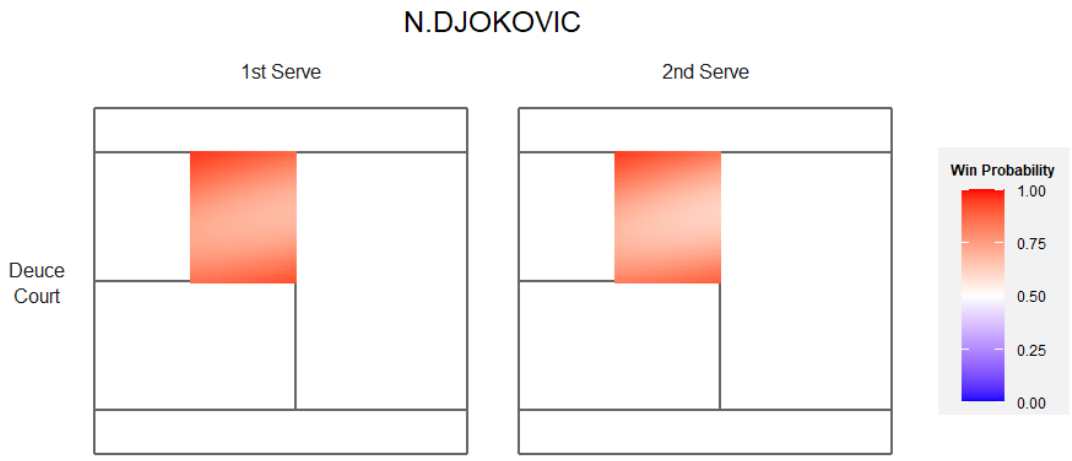


Figure 3: Estimated win probability surfaces for Djokovic's Deuce court serves. The surfaces differ across serve number due to changes in his average serve speed.

Figure 3 shows that the closer a serve lands to the lateral boundaries of the service box, the more likely it is to result in a point for the server. This is intuitive as it increases the distance the returner must travel to intercept the ball. Note that the reward surface tells us the value of a serve's *landing* location rather than the value of an *aiming* location. While the two are related, the latter additionally depends on a player's execution error.

3 Execution Error

We now turn our attention to the "risk" portion of the risk vs reward tradeoff. Aiming along a service box boundary will lead to faults about half the time, which is usually not justified by the potential payoff. The optimal location for a player to aim at depends on their distribution of serves around an intended target, which we model in this section.

We model serve bounce locations $(z_i = (x_i, y_i))^\top$ for a specific player ($p \in \{1, ..., P\}$), serve number ($s \in \{1, 2\}$), and court side ($c \in \{1 = \text{"Deuce"}, 2 = \text{"Ad"}\}$) as a bivariate Gaussian Mixture Model (GMM) with three components:

$$f(z_i) = \sum_{d=1}^{3} \theta_{p,s,c,d} \cdot \mathcal{N}_2\big(z_i \mid \boldsymbol{\mu}_{p,s,c,d},\, \Sigma_{p,s,c,d}\big)$$

where the mixture weights $\boldsymbol{\theta}_{p,s,c} = (\theta_{p,s,c,1},\, \theta_{p,s,c,2},\, \theta_{p,s,c,3})$ are a simplex representing the proportion of serves directed to each cluster. We constrain the location parameters μ such that the three components correspond to the T, Body, and Wide serving strategies. A similar model is used in Ludwig et al. [2025] to represent the distribution of baseball pitches. The mean vector $\boldsymbol{\mu}_{p,s,c,d} = (\mu^x_{p,s,c,d},\, \mu^y_{p,s,c,d})^\top$ denotes the center of each distribution. We parameterize the covariance matrices as

$$\Sigma_{p,s,c,d} = \begin{pmatrix} \tau^x_{p,s,c,d} & 0 \\ \rho_{p,s,c,d}\,\tau^y_{p,s,c,d} & \sqrt{1-\rho^2_{p,s,c,d}}\,\tau^y_{p,s,c,d} \end{pmatrix} \begin{pmatrix} \tau^x_{p,s,c,d} & 0 \\ \rho_{p,s,c,d}\,\tau^y_{p,s,c,d} & \sqrt{1-\rho^2_{p,s,c,d}}\,\tau^y_{p,s,c,d} \end{pmatrix}^\top$$

where $\tau^x_{p,s,c,d} > 0$ and $\tau^y_{p,s,c,d} > 0$ are the marginal standard deviations in the x and y directions, and $\rho_{p,s,c,d} \in (-1, 1)$ is the correlation.

Another feature of the data we must account for is a unique left censoring bound on each distribution induced by the net. Serves that would've landed to the left of these bounds were not observed due to being caught by the net. Their contribution to the likelihood is:

$$p(\text{net}_j \mid p, s, c) = \sum_{d=1}^{3} \theta_{p,s,c,d} \cdot \Phi\big(t_{p,s,c,d} \mid \mu^x_{p,s,c,d},\, \tau^x_{p,s,c,d}\big)$$

where $t_{p,s,c,d}$ is a censoring threshold and Φ denotes the univariate normal CDF.

The last significant feature of our chosen model is the use of a Bayesian hierarchy. We will pool information about all parameters used across players, meaning each player will shrink to some global average player (g) in the prior. Each player will therefore represent a compromise between their own individuality (stronger with more player-specific data) and population level trends (more influential with less player-specific data). Beyond this hierarchical pooling, the global parameters for the average player will be assigned relatively weak priors. The prior specification along with some model constraints are given below.

Location

$$\mu^x_{p,s,c,d} \sim \mathcal{N}\big(\mu^x_{g,s,c,d},\, \sigma_\mu\big), \quad \mu^x_{g,s,c,d} \sim \mathcal{N}(6,\, 1.5), \quad \mu^y_{p,s,c,d} \sim \mathcal{N}\Big(\mu^y_{g,s,c,d},\, \sigma_\mu\Big), \quad \mu^y_{g,s,c,d} \sim \mathcal{N}(0,\, 2.5)$$

$$\mu^y_{p,s,c,1} < \mu^y_{p,s,c,2} < \mu^y_{p,s,c,3}, \quad \mu^x_{p,s,c,2} \in [4.5, 6], \quad |\mu^y_{p,s,c,2}| \in [2, 2.2]$$

Scale

$$\tau^x_{p,s,c,d} \sim \mathcal{N}\left(\tau^x_{g,s,c,d},\, \sigma_\tau\right), \quad \tau^x_{g,s,c,d} \sim \mathcal{N}(0,\, 2.5), \quad \tau^y_{p,s,c,d} \sim \mathcal{N}\left(\tau^y_{g,s,c,d},\, \sigma_\tau\right), \quad \tau^y_{g,s,c,d} \sim \mathcal{N}(0,\, 2.5)$$

$$\tau^x_{p,s,c,2} \in [0, 1], \quad \tau^y_{p,s,c,2} \in [0, 0.3]$$

Correlation

$$\rho_{p,s,c,d} \sim \mathcal{N}(\rho_{g,s,c,d},\, \sigma_\rho), \quad \rho_{g,s,c,d} \sim \mathcal{N}(0,\, 0.5), \quad \rho_{p,s,c,1} < \rho_{p,s,c,2} < \rho_{p,s,c,3}$$

Censoring

$$t_{p,s,c,d} \sim \mathcal{N}(t_{g,s,c,d},\, \sigma_t), \quad t_{g,s,c,d} \sim \mathcal{N}(3,\, 1), \quad t_{p,s,c,d} \in \left[0,\, t^{\max}_{p,s,c,d}\right]$$

Hyperpriors

$$\sigma_\mu \sim \mathcal{N}^+(0,\, 2.5), \quad \sigma_\tau \sim \mathcal{N}^+(0,\, 2.5), \quad \sigma_\rho \sim \mathcal{N}^+(0,\, 1), \quad \sigma_t \sim \mathcal{N}^+(0,\, 1)$$

The ordering constraint on $\mu^y_{p,s,c,d}$ is included to avoid label switching. $\rho_{p,s,c,d}$ has a similar constraint which is justified by the geometry of serves. There are additional constraints for the body serve distribution (d = 2) as there are too few data points to inform a distribution that appeared realistic. Lastly, $t_{p,s,c,d}$ is bounded above by $t^{max}_{p,s,c,d}$, which represents the minimum observed x-coordinate for that player, serve number, and court side. It would be a contradiction for a left-censoring bound to lie to the right of any observed data points.

We visualize the model fit by extracting the posterior means of the parameters and forming 95% prediction regions, as shown by the ellipses in Figure 4. The observed bounce locations are overlaid and are colored according to the most likely serve strategy it belongs to based on the fitted model.

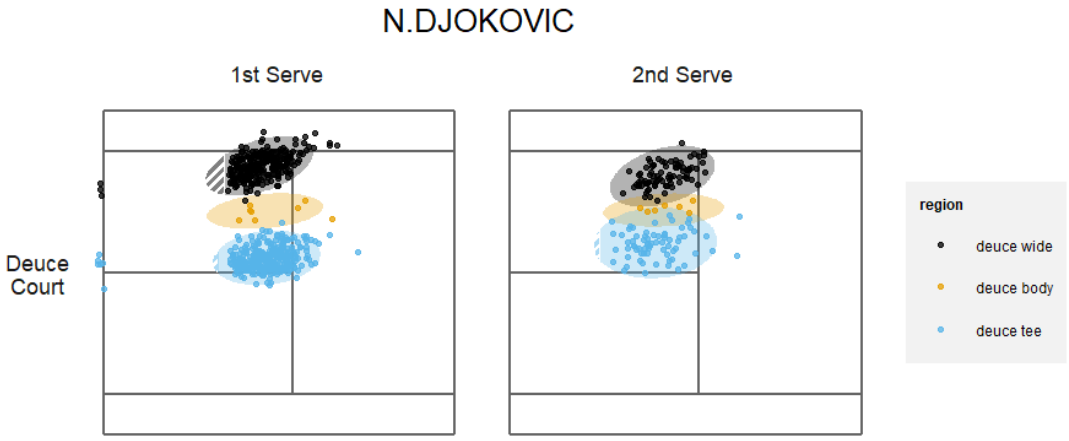


Figure 4: Ellipses represent 95% prediction regions for Djokovic's Deuce court serves. The net-censored area of each distribution is denoted by a striped pattern. Data points are classified according to the most likely serve strategy from the fitted GMM.

For each serving player, serve period, and strategy, we now have estimates of intended targets with μ, and execution error with Σ.

4 Optimization

We have now analyzed and quantified both parts of the risk vs reward tradeoff players face when deciding where to place their serves. The reward surface pushes servers to aim as close as possible to the singles sideline and center service line as this maximizes the distance the returner must run and the likelihood of winning the point. On the other hand, our model for serve distributions reveals each players execution error, telling us that since none of them can execute their serve perfectly, they each must hedge in by different amounts from the aforementioned boundaries. We now outline the process of finding optimal targets that balance this tradeoff. Full details of this process can be found in Sandholtz et al. [2026].

For each player p, serve number s, and court side c, we seek both a "Wide" target and a "T" target that maximize the server's expected point value should they aim there. We begin with the 2nd serve and then use the results to inform the 1st serve.

For a potential target, we calculate the expected point value of aiming there by weighting expected point outcomes derived in the Reward Surface section according to densities tied to the distribution of serves around the potential target. To account for the risk of faulting, areas outside the service box as well as within the net-censored region are assigned an appropriate fault penalty as the expected bounce value. For 2nd serve, that penalty is -1, indicating no chance that the server wins the point.

Unless the potential target is one of the current, "observed" targets that were estimated in the Execution Error section, we don't actually know the distribution of serves around said target. However, we can estimate the distribution through interpolation. For a potential target, we model locations the ball might land as a bivariate gaussian distribution with μ centered at the target, and Σ estimated by interpolating between the observed parameters for the "Wide" cluster and those for the "T" cluster.

After calculating the expected reward of aiming at many possible locations, accounting for execution error, we can estimate optimal targets by finding the local maxima. With this process, we found optimal 2nd serve targets for each player and region. For 1st serve, the only difference is that the penalty for faulting depends on the outcome for 2nd serve. We chose to set the penalty equal to the expected point value of aiming at one of the intended 2nd serve targets identified in the Execution Error section (whichever has a higher value between "Wide" and "T"). Another option would be to assume an optimal 2nd serve, and set that value as the penalty for faulting on 1st serve.

Figure 5 depicts a surface of the expected point values of aiming at different locations for Djokovic. The optimal "Wide" and "T" targets are marked. We see the effect of different fault penalties between 1st serve and 2nd and how that incentivizes riskier 1st serves. Since we have estimated both intended targets and optimal targets, we can now compare them, with our findings being discussed in the next section.

5 Results

There are two different ways we can compare the current targets with those that we prescribe. The first is by observing strategic bias. Let $\hat{\mu}$ represent the current targets where we estimated players aim according to the observed data. Let $\hat{\mu}^{OPT}$ represent our prescribed optimums for where players should be aiming. Then the 2-dimensional strategic bias can be calculated as $\hat{\mu} - \hat{\mu}^{OPT}$. This bias is visualized in Figure 6 for all players simultaneously, for both serve numbers, and all 4 regions.

The overall trend is players serving too narrow and too deep, which corresponds to more conservative targeting. According to our model, these players are precise enough to be able to afford taking riskier serves. This would mean aiming closer to the singles sideline for "Wide" serves and

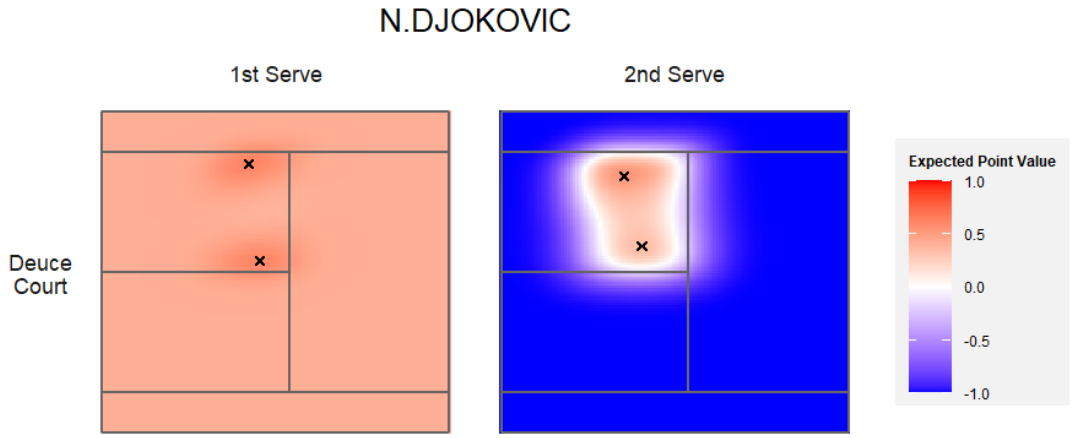


Figure 5: Surface of the expected point value of aiming at each location rather than merely landing a serve there (accounts for execution error). Local maximums are marked. The difference in fault penalties across serve numbers incentivizes riskier 1st serves and more conservative 2nd serves.

the center service line for "T" serves. We would also suggest aiming shallower even though it risks more serves getting caught by the net. However, in order to make these suggestions, we must first validate our prescribed optimums, which we do next.

The second way we can compare current and optimal targets is by showing whether there is an association between aiming closer to our prescribed optimums and winning more points. We define the following model. Let i index a (server p, returner q, serve number s, region r) cell. We model:

$$W_i \sim \text{Binomial}(n_i,\, p_i)$$

$$\text{logit}(p_i) = \alpha_{p(i)} + \beta_{q(i)} + \gamma_{r(i)} + \delta \cdot s_i + \left(\eta + \zeta_{r(i)} + \xi \cdot s_i\right) d_i$$

where:

- W_i: Number of points won by the server for the i'th cell
- α_p: server fixed effect
- β_q: returner fixed effect
- γ_r: region main effect
- δ: main effect of serve number ($s \in \{1, 2\}$, treated as numeric)
- d_i: Euclidean distance between the server's posterior mean aim point and their optimal aim point for that serve number, court side, and direction
- $\eta + \zeta_r + \xi \cdot s$: total effect of d_i for region r and serve number s (interaction)

For each of the 8 combinations of region and serve number, we test the one-sided hypothesis

$$H_0 :\ \eta + \zeta_r + \xi \cdot s \geq 0 \qquad \text{vs.} \qquad H_A :\ \eta + \zeta_r + \xi \cdot s < 0$$

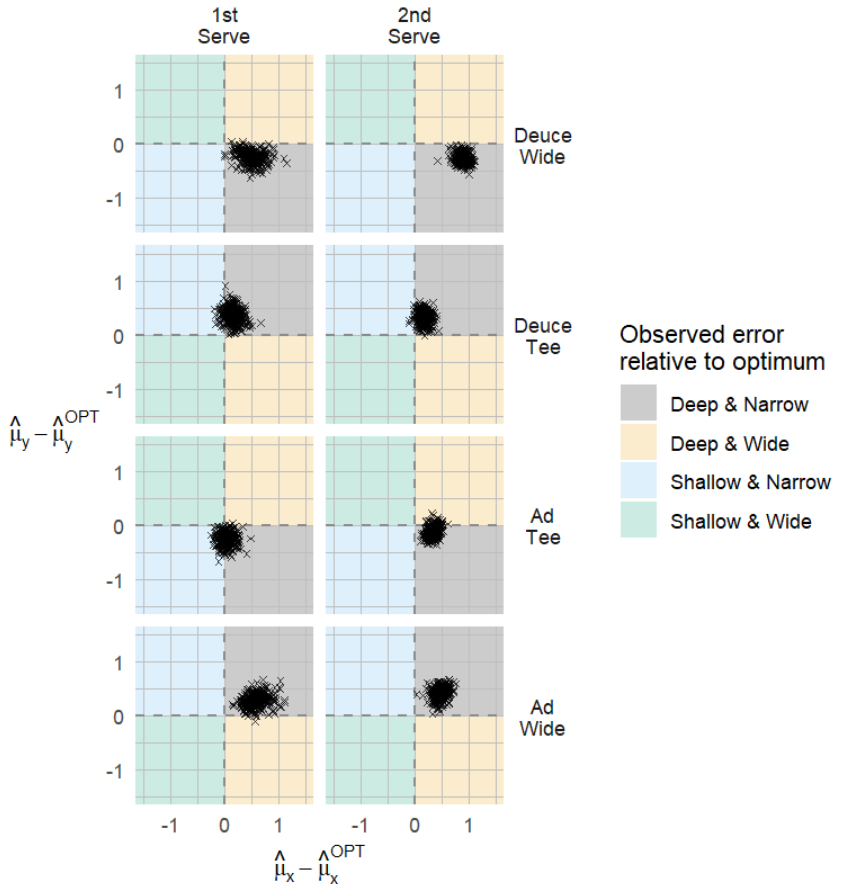


Figure 6: Strategic bias of all servers (current target - optimal target). In general, players should be aiming closer to the net and either the singles sideline for "Wide" serves or the center service line for "T" serves.

that aiming farther from the optimum reduces the server's win probability. The hypotheses can be simultaneously tested by performing a General Linear Hypothesis Test. Table 1 contains the results of this hypothesis test, showing the effect of distance from our prescribed optimum on the log-odds of winning the point for each serve number and region combination. Most of the estimates are negative, indicating aiming closer to our prescribed optimums is associated with an increased probability of winning points. However, the results are only statistically significant in the case of 1st serve Ad Wide serves and 1st serve Deuce T serves. As this project is ongoing, we will continue to look into this.

6 Conclusion

This far, we have outlined a method for attaining optimum targets to aim towards on a serve for different players and serve numbers. This process has involved attaining reward surfaces informed by point outcomes as well as estimating execution error which influences how far different players

Serve	Court Side	Direction	Estimate	Std. Error	z	p
1st	Ad	T	-0.315	0.154	-2.05	0.121
	Ad	Wide	-0.321	0.104	-3.10	0.007^{**}
	Deuce	T	-0.514	0.124	-4.15	$< 0.001^{***}$
	Deuce	Wide	-0.195	0.092	-2.13	0.101
2nd	Ad	T	-0.090	0.172	-0.52	0.839
	Ad	Wide	-0.097	0.116	-0.83	0.700
	Deuce	T	-0.289	0.126	-2.29	0.069
	Deuce	Wide	0.029	0.082	0.36	0.992

$^{**}p < 0.01$; $^{***}p < 0.001$. One-sided tests: $H_A : \eta + \zeta_r + \xi \cdot s < 0$.

Table 1: Estimated total effect of distance from optimal aim point on server win probability, by serve number, court side, and direction.

should hedge in from court boundaries. To continue this research, we will be finding posterior draws of optimal targets rather than just estimating one for each player, serve number, and region. This can help us confirm that our optimal targets are significantly different from those that are observed. We will also continue making small adjustments to models used to determine whether they can be improved.

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